

One-sided Carrying Analysis to Reduce Musculoskeletal Disorders

Project Proposal

TU Wien - ASM2 Project

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Introduction

In assembly lines, the finished product from one station often needs to be transported to the next processing step. In less automated enterprises, this task is frequently performed manually by human operators. Especially for large and heavy products, the way these loads are carried and handled can significantly affect the operator's posture and, over time, their health. One-sided load carrying is a common occurrence in assembly and logistics environments and has been associated with increased spinal loading and lower back pain [1]. Despite this, most workplaces still lack objective methods to continuously monitor carrying behavior and posture.

This project investigates whether wrist-worn IMU sensors can reliably distinguish between one-sided and two-handed carrying of objects in an assembly environment (H1).

We hypothesize that the asymmetry between both wrists is most pronounced during the walking phase, because the unloaded wrist swings more freely while the loaded wrist remains more stabilized, making one-sided carrying easier to detect than during the pick-up phase (AH1).

We further hypothesize that detection performance differs depending on the size and weight of the carried object, with larger and heavier objects leading to more distinct movement patterns than smaller and lighter objects (AH2).

Methodology

Participants will wear two wrist-mounted IMUs (one per wrist) recording 3-axis accelerometer and 3-axis gyroscope data at 100 Hz. They will carry two objects differing in size under two carrying conditions: one-sided (dominant arm only) and two-handed. Each trial consists of three phases: picking the object up, walking a short, fixed route (~10 m), and setting the object down. There will be one version where the participant picks something up from a table and one version where it is picked up from the floor. Each combination of object and carrying condition will be repeated across multiple trials per participant to ensure sufficient data coverage.

The three motion phases will be segmented from the continuous IMU stream using acceleration magnitude thresholds. Features will be extracted per phase and per wrist.

A binary classifier (symmetric vs. asymmetric carry) will be trained on these features. Classification performance will additionally be compared across the three motion phases to test the auxiliary hypothesis that the walking phase yields the strongest asymmetry signal.

Additionally, unsupervised clustering will be applied to the extracted features to explore whether natural groupings emerge in the movement data - for example, distinct postures of object-size-dependent motion profiles.

Results and Impact

The aim of this study is to detect one-sided carrying in an assembly environment, as this may pose a risk to the operators' health due to increased physical strain and unfavorable posture. Once one-sided carrying is detected, this information can be used to provide feedback to operators about their carrying behavior and posture. If necessary, appropriate ergonomic interventions, such as training, workplace adjustments, or the use of exoskeletons, can be introduced to reduce physical stress and support healthier working postures.

References

- [1] et al. Luo. Neuromuscular and kinetic adaptations to symmetric and asymmetric load carriage during walking in individuals with chronic low back pain. Bioengineering, 13(1):82, 2025.